Cornell University

# Bayesian Time-Series Structural Decomposition of Crude Oil Market Dynamics

with Applications to U.S. Macroeconomic Forecasting

Tony Au, Merve Abacı

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# 1 Introduction: Background, Literature, and Assumptions

The goal of the research is to study **how unexpected shock** in the crude oil market — driven by supply, global industrial commodities demand, and precautionary demand — **affect the U.S. economy.**. The effects of oil prices on macroeconomic determinants have been long discussed in the literature, especially after the recession in the 1970s, which coincided with oil supply crisis followed by oil embargo (Kilian, 2001). Many of those models consider oil supply shocks as an exogenous variable that is affected by political disturbances, while keeping all the other variables constant (Kilian, 2009). However, this approach has two fundamental problems. Firstly, macroeconomic variables tend to affect oil prices, which creates reverse causality, and the direction of the effects are no longer well defined. Even when an oil-price shock seemed to be affected by a political event such as an oil embargo, the success of cartels highly depends on the overall strength of the U.S. dollar, invalidating the exogeneity assumption (Kilian, 2004). Secondly, they do not differentiate between distinct types of shocks, which have different effects on macroeconomic outcomes. Furthermore, demand-related shocks may also have an impact on the overall economy, violating the ceteris paribus assumption.

This paper is inspired by a structural VAR model that is proposed by Kilian (2009). Three types of shocks are considered: 1) oil supply shocks driven by the physical availability of crude oil, 2) aggregate demand shocks driven by fluctuations in the global business cycle, and 3) precautionary demand shocks driven by market concerns about the future availability of oil supplies. Accordingly, these three shocks have different long-term and short-term effects, and their relevance comes from the need for different policy considerations. This paper extends the original research in two ways. Firstly, we extended the time frame using the most recent available data. Secondly, we introduce a Bayesian method instead of a frequentist method for uncertainty interpretation. We will verify the result proposed by the Kilian (2009).

# 2 Data and Variable Construction

The model uses three monthly endogenous variables in following ordering, suggested by Kilian (2009): the percent change in global crude oil production ($\Delta prod$), the real economic activity index ($rea$), and the log real price of crude oil ($rpo$).[1]

The production series is built from levels of world crude oil output. We first difference production in logs and scale the monthly change to an annualized percentage,

$$\Delta prod_t = 1200 \ln\left(\frac{prod_t}{prod_{t-1}}\right), \qquad rpo_t = 100 \ln\left(\frac{P_t}{CPI_t}\right), \tag{1}$$

where $prod_t$ is global crude oil production (thousands of barrels per day). Thus, $\Delta prod_t$ is not used in levels; differencing and the $\times 1200$ scaling convert the production level into an annualized percent change.The $rea$ series is taken directly as the Kilian real economic activity index (percent deviations from trend; 2019-corrected version for the extended sample). The $rpo$ series is constructed as formula (8), where $P_t$ is the nominal imported crude oil price and $CPI_t$ is the U.S. consumer price index.

The baseline sample follows the sample in Kilian (2009) (1973:02–2007:12) and is extended through 2025 using the same transformations. Observations from October 2025 onward are excluded because CPI for that month is unavailable (U.S. government shutdown), which prevents consistent construction of $rpo$.

[1] $\Delta prod$ is obtained from the U.S. Energy Information Administration (EIA) series for world crude oil production (including lease condensate). The nominal oil price component of $rpo$ is the EIA *Refiner Acquisition Cost of Crude Oil, Imported*; this series is deflated using the U.S. CPI (`CPIAUCSL`, Federal Reserve Economic Data). The $rea$ index is obtained from the updated Kilian real activity measure (Federal Reserve Bank of Dallas; 2019-corrected).

### Construction of the real activity variable (rea)

As a direct measure of demand fluctuation driven by business cycles is not readily available, this model uses **dry cargo single voyage ocean freight rates**, (a.k.a. Kilian index), exploiting the close relationship between demand for transportation and real business activity. The index is constructed by calculating the changes in freight rates deflated by U.S. CPI and taking the logarithm, refer to (Kilian, 2009) for more details. The data set is available in the monthly report of "Shipping Statistics and Economics" published by Drewry Shipping Consultants Ltd.

The index is motivated by the following mechanism: Supply of shipping is relatively flat in the short run, and a small increase in demand does not generate large price increases, as there is excess capacity that is not utilized. However, the supply of ships is not able to respond to a greater increase in demand. When the cargo ships reach full capacity, the slope of the supply will become increasingly steeper, finally reaching a fully vertical inelastic supply, where additional increases in freight rate do not increase the supply. The freight rates drops only in the long run, after additional shipbuilding. The index captures the shifts in the demand for industrial commodities in global business markets, not just oil, and it can be used as an indicator of strong cumulative global demand pressures.

### Data Exploration


Figure 1: Time series plot of the data: 1973–2025

In Figure 8, greater volatility in the real activity index after 2008 crisis is visible. We also observe a downward movement in oil prices around the same years. In the top panel, a sharp decline in oil production in 2020 is evident due to the COVID-19 shock. Hence, the extended time frame exhibits greater variability, particularly for real activity and oil prices.

| | series | adf_stat | adf_pvalue | kpss_stat | kpss_pvalue |
|---|---|---|---|---|---|
| **0** | d_prod | -6.026682 | 1.450329e-07 | 0.036947 | 0.10 |
| **1** | rea | -4.417180 | 2.767273e-04 | 0.204174 | 0.10 |
| **2** | rpo | -1.505072 | 5.310006e-01 | 2.619633 | 0.01 |

Figure 2: Stationary Checks: ADF and KPSS

The ADF and KPSS tests provide evidence for d-prod and rea are stationary, as the null hypothesis of unit root is rejected by ADF, while KPSS does not reject stationarity. However, rpo seems to be non-stationary, which is confirmed by the results of both tests.

## Model Assumptions and Identification

For this paper, we follow the same assumptions proposed by Kilian (2009). Oil supply shocks are defined as unpredictable innovations in global oil production that do not respond to innovations in the demand for oil **within the same month**. Oil-producing countries are slow to respond to demand shocks, as there is uncertainty about crude oil market and adjustments are costly. Oil producers set production levels based on expected demand trends and do not readjust as high-frequency variation of demand is noisy.

Kilian argued that the demand shock can be decomposed into 1) aggregate demand, which is intended to capture global demand of the commodity market captured by (rea); 2) oil-specific demand, where he suggested it is equal to oil precautionary demand shock.

We use the terms oil-specific demand shock and oil precautionary demand shock interchangeably throughout the paper. In principle, oil-market-specific demand shocks may capture omitted factors, such as particularly colder winters. However, by construction, the model ensures that these shocks are orthogonal to both crude oil supply shocks and global demand shocks for industrial commodities. Furthermore, there is little evidence that the oil-specific demand shocks are associated with factors such as unusually cold winters, since such effects are likely to average out at the global level. Instead, these shocks primarily reflect precautionary demand driven by uncertainty about future oil supply shortfalls. First, the timing of the estimated shocks closely coincides with major geopolitical events, such as the Persian Gulf War, that increased concerns regarding future oil availability. Second, the overshooting of oil prices following these shocks is consistent with theoretical models of precautionary demand under increased uncertainty. Finally, Kilian (2009) shows that movements in the real price of oil associated with these shocks are highly correlated with independent measures of precautionary demand constructed from oil futures prices.

The model also assumes that increases in the real price of oil caused by precautionary shocks will reduce global economic activity not immediately, but with a lag of at least one month. This reflects sluggish adjustment in the real economy and firms do not instantly cut production when oil prices increase.

Finally, the real price of oil reacts to all three types of shocks. This assumption reflects asset and commodity prices adjust quickly. Expectations affected by precautionary demand also move the prices instantly.
Therefore, we introduce a **short-run restriction** to disentangle the oil-market shock for our interest of study.

$$e_t = \begin{pmatrix} e_t^{\Delta prod} \\ e_t^{rea} \\ e_t^{rpo} \end{pmatrix} = \begin{pmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} \epsilon_t^{\text{oil supply}} \\ \epsilon_t^{\text{aggregate demand}} \\ \epsilon_t^{\text{oil-specific demand}} \end{pmatrix} \tag{2}$$

Where $e_t$ is the vector of reduced-form VAR innovations, and $\epsilon_t$ is the vector of orthogonal structural shock, containing the variable of our interest.

## 3 Methodology

For the hyperperemeter selection and model fitting diagnostic, one can refer to appendix for details.

### 3.1 Bayesian VAR model with Normal Inverse-Wishart prior

We first estimate a Bayesian VAR in reduced form,

$$\boxed{z_t = c + \sum_{i=1}^{p} A_i z_{t-i} + e_t, \qquad e_t \sim \mathcal{N}(0, \Omega),} \tag{3}$$

where $z_t = (\Delta prod_t,\ rea_t,\ rpo_t)^T$ and lag $p = 24$ (i.e. 2 years). Let $n$ denote the number of variables, $k = 1 + np$ the number of regressors per equation, and $Z_t^T = [1,\ z_{t-1}^T,\ \ldots,\ z_{t-p}^T]$ the $1 \times k$ vector of regressors at $t$. Stacking lags, (3) can be written as

$$z_t^T = Z_t^T B + e_t^T, \tag{4}$$

where the $k \times n$ coefficient matrix $B = \{c, A_1, \ldots, A_p\}$ .

**NIW prior.** We use a weak conjugate Normal–Inverse-Wishart prior:

$$\boxed{\text{vec}(B) \mid \Omega \sim \mathcal{N}\big(\text{vec}(\underline{B}),\ \Omega \otimes \underline{V}\big), \qquad \Omega \sim \mathcal{IW}(\underline{\nu},\ \underline{S}),} \tag{5}$$

equivalently $B \mid \Omega \sim \mathcal{MN}(\underline{B}, \underline{V}, \Omega)$ (matrix-normal, not Minnesota), with prior hyperparameters

$$\underline{B} = 0_{k\times n} = 0_{73\times 3}, \quad \underline{V} = 10 \times I_k\ (k = 1 + np = 73), \quad \underline{\nu} = n + 2 = 5, \quad \underline{S} = I_n = I_3.$$

Thus coefficients are shrunk toward zero with a relatively diffuse prior ($\underline{V}^{-1} = 0.1\, I_k$), and $\Omega$ receives a minimally proper inverse-Wishart prior.

### 3.2 Markov Chain Monte Carlo: Gibbs Sampler Algorithm

Let $Y$ and $X$ denote the $T_{\text{eff}} \times n$ and $T_{\text{eff}} \times k$ data matrices whose $t$-th rows are $z_t^T$ and $Z_t^T$. We simulate the posterior by repeated sampling from two conditional posteriors.

**Initialization.** Start the chain from OLS estimates on the effective sample ($T_{\text{eff}} = T - p$):

$$B_{\text{OLS}} = (X^T X)^{-1} X^T Y, \qquad \hat{E}_{\text{OLS}} = Y - X B_{\text{OLS}}, \qquad \Omega_{\text{OLS}} = \frac{\hat{E}_{\text{OLS}}^T \hat{E}_{\text{OLS}}}{T_{\text{eff}}}. \tag{6}$$

Set the starting values to

$$B^{(0)} = B_{\text{OLS}}, \qquad \Omega^{(0)} = \Omega_{\text{OLS}}.$$

**Gibbs iterations.** For $s = 1, 2, \ldots, M$:

**Draw 1: coefficients given $\Omega^{(s-1)}$.**

$$\boxed{B^{(s)} \mid z,\ \Omega^{(s-1)} \sim \mathcal{MN}\big(\bar{B}^{(s)},\ \bar{V}^{(s)},\ \Omega^{(s-1)}\big),}$$

equivalently $\mathrm{vec}(B^{(s)}) \sim \mathcal{N}\big(\mathrm{vec}(\bar{B}^{(s)}),\, \Omega^{(s-1)} \otimes \bar{V}^{(s)}\big)$, where

$$\bar{V}^{(s)} = \left(\underline{V}^{-1} + \sum_{t=1}^{T_{\text{eff}}} Z_t Z_t^T\right)^{-1} = \left(\underline{V}^{-1} + X^T X\right)^{-1}, \tag{7}$$

$$\bar{B}^{(s)} = \bar{V}^{(s)} \left(\underline{V}^{-1}\underline{B} + \sum_{t=1}^{T_{\text{eff}}} Z_t z_t^T\right) = \bar{V}^{(s)} \left(\underline{V}^{-1}\underline{B} + X^T Y\right). \tag{8}$$

Here $z_t^T$ is the $t$-th row of $Y$ and $\widehat{z}_t^{\,T} = Z_t^T B^{(s)}$ is the fitted value.

**Draw 2: covariance given $B^{(s)}$.**

$$\boxed{\Omega^{(s)} \mid z,\, B^{(s)} \sim \mathcal{IW}\big(\bar{\nu}^{(s)},\, \bar{S}^{(s)}\big),}$$

with

$$\bar{\nu}^{(s)} = T_{\text{eff}} + \underline{\nu}, \tag{9}$$

$$\begin{aligned}\bar{S}^{(s)} &= \underline{S} + \sum_{t=1}^{T_{\text{eff}}} (z_t - B^{(s)T} Z_t)(z_t - B^{(s)T} Z_t)^T + (B^{(s)} - \underline{B})^T\, \underline{V}^{-1} (B^{(s)} - \underline{B}) \\ &= \underline{S} + (E^{(s)})^T E^{(s)} + (B^{(s)} - \underline{B})^T\, \underline{V}^{-1} (B^{(s)} - \underline{B}), \quad E^{(s)} = Y - X B^{(s)}. \end{aligned} \tag{10}$$

Discard a fair amount of initial burn-in of draws (2000 used here); retain the rest for posterior summaries of $B$ and $\Omega$ (10000 remaining).

### 3.3 Structural Vector Auto-Regressive Model (SVAR)

The SVAR representation is:

$$\boxed{A_0 z_t = a + \sum_{i=1}^{p} A_i z_{i-1} + \epsilon_t} \tag{11}$$

Where $z_t = (\Delta prod_t, rea_t, rpo_t)^T$ and $e_t = A_0^{-1}\epsilon_t$ can be decomposed in a recursive structure as equation (2).

The short-run restriction structure can be obtained by using a Cholensky decomposition:

$$e_t = A_0^{-1}\epsilon_t, \quad \epsilon_t \sim N(0, I)$$

With covariance matrix:

$$\Sigma = A_0^{-1}(A_0^{-1})^T$$

Therefore, we decomposed independent structural shock from equation (11) using the recursive structure.

### 3.4 IRFs and FEVD

Under a NIW prior, Gibbs sampling yields draws $\{B^{(d)}, \Sigma^{(d)}\}_{d=1}^{D}$

**(1) Companion form.** Stack lags as $x_t = (z_t', z_{t-1}', \ldots, z_{t-p+1}')'$ so that

$$x_t = \mathcal{A}^{(d)} x_{t-1} + \mathcal{B} u_t, \qquad z_t = \mathcal{J} x_t, \qquad \mathcal{J} = [I_n \; 0 \; \cdots \; 0],$$

where $\mathcal{A}^{(d)}$ is built from the lag blocks in $B^{(d)}$ (intercept in the first row of $B^{(d)}$).

**(2) MA($\infty$) representation.** With recursive identification $u_t = C^{(d)}\varepsilon_t$, $\mathbb{E}[\varepsilon_t\varepsilon_t'] = I_n$, and $C^{(d)} = \text{chol}(\Sigma^{(d)})'$,

$$z_t = \mu + \sum_{h=0}^{\infty} \Psi_h^{(d)} \varepsilon_{t-h}, \qquad \Psi_h^{(d)} = \mathcal{J}(\mathcal{A}^{(d)})^h \mathcal{J}' C^{(d)}, \quad h = 0, 1, \ldots, H = 15,$$

**(3) IRFs and FEVD.** The orthogonalized impulse response is the partial effect of a unit structural shock:

$$\boxed{\psi_{ij,h}^{(d)} = \frac{\partial z_{i,t+h}}{\partial \varepsilon_{j,t}}\Big|_{\varepsilon_t=0} = \left[\Psi_h^{(d)}\right]_{ij}.}$$

The forecast error variance share is

$$\boxed{\theta_{ij}^{(d)}(h) = \frac{\sum_{s=0}^{h}\left(\psi_{ij,s}^{(d)}\right)^2}{\sum_{k=1}^{n}\sum_{s=0}^{h}\left(\psi_{ik,s}^{(d)}\right)^2}, \qquad \sum_{j=1}^{n} \theta_{ij}^{(d)}(h) = 1,}$$

reported as $\text{FEVD}_{ij}^{(d)}(h) = 100\, \theta_{ij}^{(d)}(h)$.

**(4) Bayesian 68% & 95% credible intervals.** For each $(i, j, h)$, summarize $\{\psi_{ij,h}^{(d)}\}_{d=1}^{D}$ and $\{\theta_{ij}^{(d)}(h)\}_{d=1}^{D}$ by the posterior median and equal-tailed intervals at $(16\%, 84\%)$ and $(2.5\%, 97.5\%)$.

### 3.5 Shock-Augmented Regression Model for U.S. Economy

The quarterly shocks are constructed by averaging the monthly structural innovations:

$$\hat{\zeta}_{jt} = \frac{1}{3}\sum_{i=1}^{3} \hat{\varepsilon}_{j,t,i}, \quad j = 1, 2, 3 \tag{12}$$

where $\hat{\varepsilon}_{j,t,i}$ refers to the estimated residual for the $j$th structural shock in the $i$th month of the $t$th quarter of the sample.

The effects on U.S. macroeconomic aggregates are examined based on the regressions:

$$\Delta y_t = \alpha_j + \sum_{i=0}^{12} \phi_{ji}\hat{\zeta}_{jt-i} + u_{jt}, \quad j = 1, 2, 3 \tag{13}$$

and

$$\pi_t = \delta_j + \sum_{i=0}^{12} \psi_{ji}\hat{\zeta}_{jt-i} + v_{jt}, \quad j = 1, 2, 3 \tag{14}$$

where $u_{jt}$ and $v_{jt}$ are potentially serially correlated errors.

Since real GDP is unavailable at monthly frequency, Kilian (2009) constructed quarterly shock measures by averaging monthly structural innovations. Under the identifying assumption that quarterly shocks are predetermined with respect to U.S. real GDP growth and inflation, the paper estimates their effects using distributed lag regressions with up to 12 quarterly lags.

# 4 Empirical Result and Discussion

## 4.1 Impulse Response Function (IRFs)

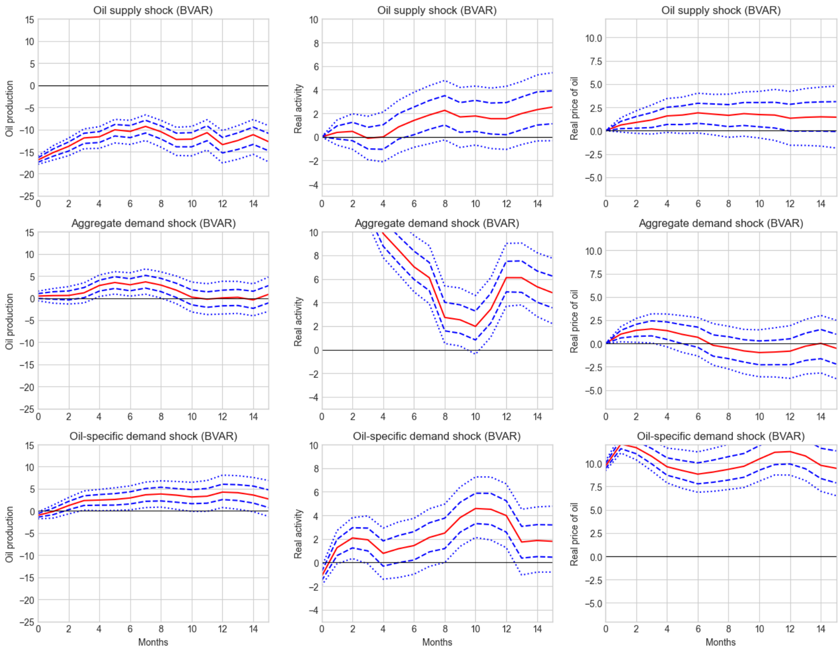


Figure 3: IRFs and credible intervals (68%, 95%) from posterior draws

**Row 1: Oil supply shock** An oil supply shock lowers oil production immediately and persistently: the median response is strongly negative on impact and remains below zero for the full 15-month horizon, with only a slight upward drift over time. The median falls from roughly 17 at month 0 to about 12 by month 15. The 95% credible interval stays entirely below zero, so this effect is robustly negative.

For real activity (dry cargo single-voyage ocean freight rates), the median response is near zero at first, then turns positive after about month 4 and rises gradually to roughly 2–2.5 by month 15. The 68% credible interval lies above zero from about month 6 onward; the 95% interval includes zero for the entire horizon, so the positive effect is suggestive but not fully robust at the 95

For movements in the real price of oil, the median starts at zero and increases gradually, peaking at roughly 2 and remaining positive through month 15. The 68% credible interval is above zero from about month 1 through month 15; the 95% interval includes zero throughout, so significance is again clearest at the 68% level.

**Row 2: Aggregate demand shock** An aggregate demand shock has little effect on oil production at impact (zero by identification) and only a small, delayed positive response afterward. The median is near zero for most of the horizon, with a modest rise between roughly months 3–10 (peak well below 5). Credible intervals largely overlap zero, which is consistent with Kilian's assumption that oil production cannot adjust meaningfully within the same month.

The effect on real activity is large, immediate, and persistent. The median jumps above 10 on impact, falls to about 2 around month 10, then recovers to roughly 5 by month 15. The 95% credible interval remains above zero for the full horizon, indicating a strong and statistically robust positive response.

For the real price of oil responses, the median is weakly positive early on, peaks at about 1.5 around month 3, and then declines, crossing below zero around month 7. The 68% credible interval is above zero only for roughly months 1–6; thereafter the response is around or below zero. This pattern is only a temporary positive effect, not a sustained increase in oil prices.

**Row 3: Oil-specific (precautionary) demand shock** An oil-precautionary demand shock raises oil production after a short delay. The median is zero at impact by construction, turns positive from about month 2, and stabilizes around 3–4 from roughly month 6 onward. The 68% credible interval is above zero from about month 2; the 95% interval is above zero from about month 4, so the production response becomes robust over time.

The response of real activity is positive but not smooth: the median is slightly negative at month 0, rises to about 2 around month 2, dips in the middle of the sample, and peaks near 4–4.5 around month 10 before easing. The 68% credible interval is mostly above zero after month 1, but the 95% interval crosses zero around months 4–7 and again after about month 13. The positive effect on activity is therefore clearest in the short run and again in the middle of the horizon, rather than uniformly across all months.

For movements in the real price of oil, the median response is large and immediate (about 10 on impact), peaks near 12 around month 2, and remains around 10 through month 15. The 95% credible interval stays well above zero for the entire horizon, making this the strongest and most persistent result in the figure.

**Takeaway** Overall, the extended-sample results are broadly in line with Kilian's (2009) identification assumptions and main qualitative conclusions, but they are not entirely consistent with his reported magnitudes and significance patterns.

First, oil-specific demand shocks are the **dominant driver of movements in the real price of oil**: the response is large, immediate, and significant even at the 95% level for the full horizon. That supports Kilian's emphasis on precautionary demand for oil-price dynamics.

Second, oil production behaves largely as in Kilian's framework for aggregate demand (small, delayed, mostly insignificant effect in row 2, column 1). For oil-specific demand, however, production does rise significantly in the extended sample (row 3, column 1)—a point that should not be summarized as insignificant for both demand shocks, which differs from Kilian's finding that both demand shocks leave production largely unchanged.

Third, relative to Kilian, we find some positive responses of real activity and real price of oil to oil supply shocks (row 1, columns 2–3), whereas he treated those responses as insignificant; in our plots, those effects are mainly significant at the 68% level, with 95% bands often including zero. We also do not find a sustained increase in real price of oil responses to aggregate demand shocks after about month 6 (row 2, column 3), whereas Kilian emphasized a longer-lasting positive price effect.

In summary, identification and timing restrictions still look reasonable, precautionary (oil-specific) demand remains the key driver of oil-price movements, but the extended sample shows stronger and sometimes more persistent responses than Kilian's original results in a few panels—especially oil production after oil-specific demand and early positive activity/price responses to oil supply shocks—while aggregate demand's effect on oil prices is weaker and more short-lived than in his baseline.

### 4.2 Forecast Error Variance Decomposition (FEVD)

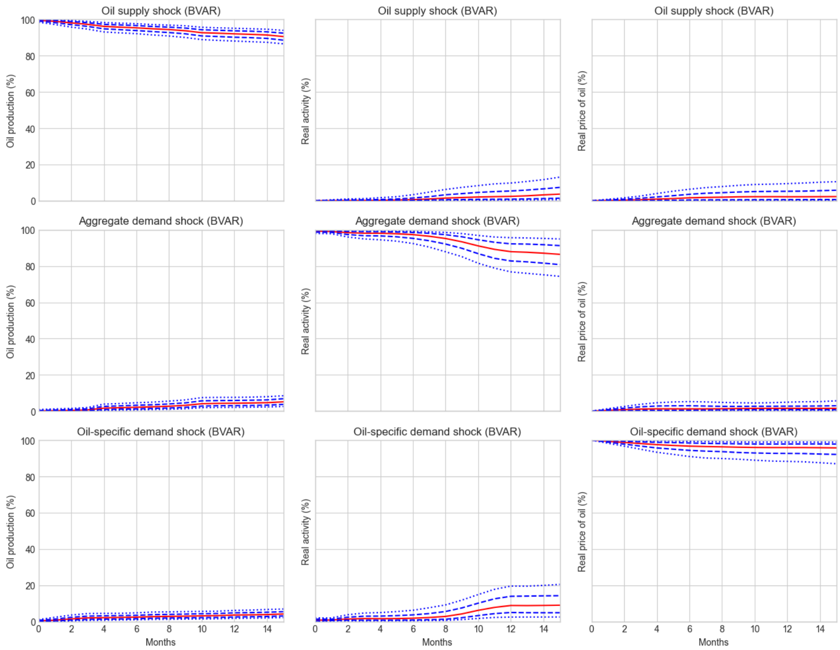


Figure 4: FEVD

Table 1: Diagonal forecast error variance shares at horizon $h = 15$ (posterior median, percent).

| Variable | Own-shock FEVD at $h = 15$ (%) |
|---|---|
| oil production (d_prod) | 90.5 |
| real activity (rea) | 86.5 |
| real price of oil (rpo) | 95.7 |

The plot and table suggested that most of the forecast error can be explained by the diagonal variable, which is consistent with Kilian's finding and assumption. He proposed the short-run restriction by arguing that oil production change shock is mainly driven by oil supply shock as the adjustment of oil production is highly inelastic, it is consistent with the first column of plots that the two demand shock do not explain much of the forecast error. The third column of plots agree with his result of oil precautionary shock is the main driver for the real price of oil with up to 1 FEVD at the beginning and $\approx 96\%$ FEVD at horizon 15.

The second columns of the plots suggests that real activity forecast error is mostly explained by aggregate demand shock with close to 100% FEVD at the beginning, but the dominant relationship is weaken over time and drops to $\approx 87\%$, suggesting that other two structural shocks helped explain some of the variance over time.

## 4.3 Cumulative Effect Decomposition on Crude Oil Real Price

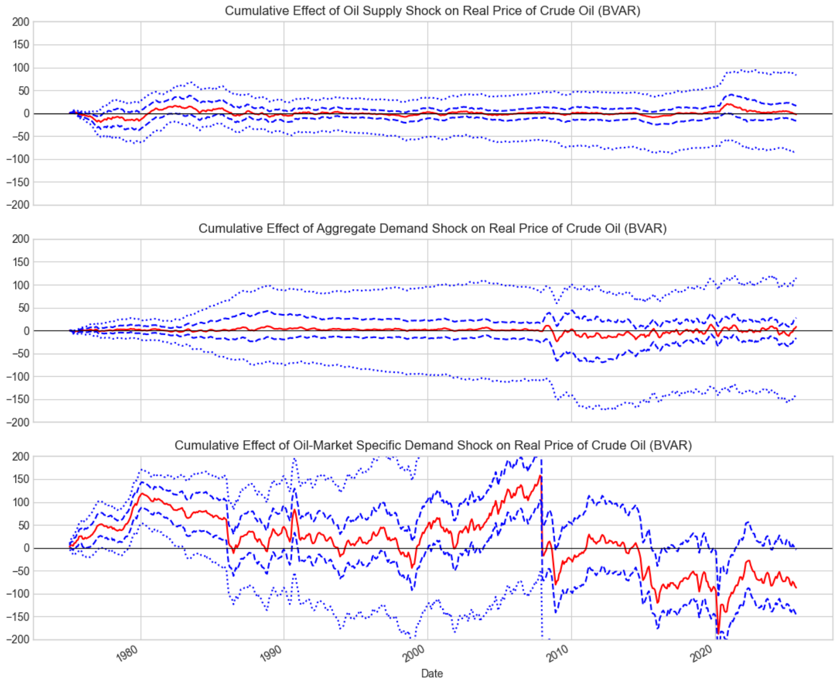


Figure 5: Cumulative Effect Decomposition on Crude Oil real Price

The first two figures show that the cumulative effects of the oil supply shock and the aggregate demand shock are insignificant: the 95% credible bands cover zero throughout. The aggregate demand shock has a wider interval, suggesting greater uncertainty about its cumulative effect on the real price of oil.

The third figure shows a statistically significant cumulative effect from the oil-market-specific (precautionary) demand shock through about 2008, after which the point estimate turns negative. During the significant phase, the cumulative contribution is positive and the credible intervals often exclude zero—notably around the late 1970s and early 1980s and again in the run-up to 2008. These episodes align with major historical disruptions in oil markets, including the 1979 Iranian Revolution and the 1980–81 Iran–Iraq War, the 2008–09 global financial crisis, and, more recently, the 2022 Russia–Ukraine war (see Appendix 6.5 for a richer summary). The credible intervals in this panel are also much wider than in the other two—roughly twice as wide at some points—so inference on the magnitude of the effect is less precise even when significance is clear.

The takeaway is that, among the shocks considered in the BVAR, oil precautionary (oil-market-specific) demand shocks appear to be the dominant driver of cumulative movements in the real price of crude oil in the pre-2008 period, when their cumulative impact is significant and positive. After 2008, the sign reversal indicates that the same shock contributes negatively on a cumulative basis in later years, while uncertainty about its magnitude remains high throughout. One possible interpretation is that this shock captures precautionary storage and demand, shifts in risk and expectations, and news-driven changes in desired inventories—forces that can move prices sharply but are hard to pin down econometrically, which is consistent with the wide credible bands.

### 4.4 Regression Model: Shock Decomposition on U.S. GDP and CPI

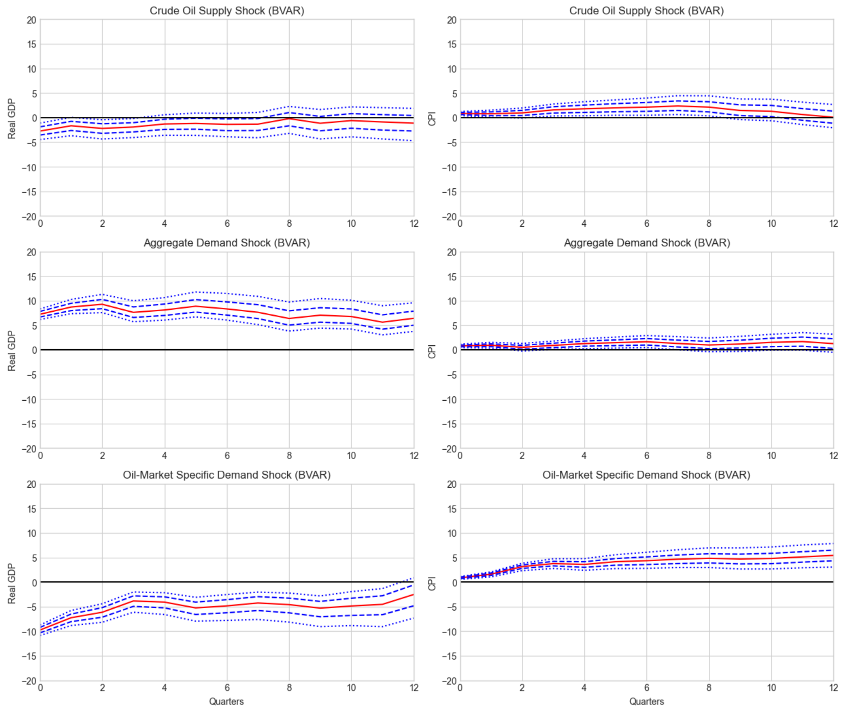


Figure 6: Responses of U.S. real GDP and CPI level to each structural shock

The key findings reveal that oil supply disruptions cause temporary GDP declines that is significant only for the initial period, while having relatively limited and positive effects on the consumer price level. This suggest that short-run oil supply shortages are not enough to generate strong and persistent macroeconomic disruptions. Aggregate demand shocks initially stimulate the economy, indicating a stronger global demand and expansion, but become recessionary with a delay as higher commodity prices dominate.

Finally, precautionary demand shocks lower real GDP while raising consumer prices, with strong and significant effects that persist. These results explain why the post-2003 oil price surge (driven primarily by aggregate demand) did not immediately trigger a major U.S. recession, unlike earlier oil shocks driven by different underlying causes. This suggests that expectations regarding the availability of crude oil may have a stronger effect on macroeconomic outcomes compared to the actual shocks related to physical availability.

## 5 Conclusion

Our analysis is consistent with Kilian's (2009) results when we use the full sample from 1973 to 2025. More specifically, oil supply shocks have limited macroeconomic effects, while demand shocks play a more important role in explaining the U.S. macroeconomic trends and oil price markets. However, when we use the extended sample, starting from the year 2007, the model revealed important differences, showing increased volatility and different effects for structural shocks.

Particularly, oil supply shocks have an instant, persistent, and negative effect on oil production, but have a weak positive effect on real activity and real price of oil. Aggregate demand shocks positively and weakly affect oil production, with the IRF staying close to zero except for the mid-term effects. They also have an initially positive and weak impact on the real price, which becomes negative in the long-run. The last observation about the aggregate demand is that they have a strong, persistent, and positive effect on real activity. Finally, oil-specific demand shocks have a weak and positive effect both on oil production and real activity, but they significantly and positively affect real price of oil.

For macroeconomic variables, the shock decomposition suggests that oil supply shocks cause temporary and negative effects on real GDP and positively affect CPI inflation; however, both relations seem to be weak, especially with a low magnitude for CPI. Aggregate demand positively and strongly affects real GDP, while it has positive effects on CPI inflation. While demand shocks initially stimulate the economy, in the long run, price effects take offer and the shock becomes recessionary. Finally, oil-market-specific shocks affect real GDP positively and CPI inflation negatively, while both effects seem to be strong and significant.

The weak off-diagonal FEVD suggests the joint-estimate model that we used does not increase much interpretability across variables, as each of the forecast error variances is mostly explained by its own shocks rather than cross-variable effects. FEVD graphs suggests that oil production mostly effected but oil supply shocks, whereas real activity is affected by aggregate demand shocks, and real price is affected by oil-specific demand shocks. These results are consistent with Kilian's assumptions, but they may indicate some weakness in the recursive SVAR assumptions in the extended sample. However, a multivariate system might still be preferable due to the aforementioned justifications.

One possible limitation of our analysis is that the data quality. For the new Kilian index, flight rates, were corrected after 2019, which could have caused breaks in the extended data. Furthermore, the data are only extended through October 2025 because the CPI for that month was unrecorded due to the government shutdown, further shortening our estimation timeline. Finally, the hyper contango in oil production in 2020 (negative oil price), due to COVID-19, may cause additional volatility affecting the stationarity of the model.

# 6 References


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# 7 Appendix

## 7.1 Model Selection

| Lag | AIC | Lag | AIC | Lag | AIC |
|---|---|---|---|---|---|
| 1 | 16.098686 | 13 | 15.960102 | 25 | 16.096279 |
| 2 | 15.947494 | 14 | 15.979509 | 26 | 16.109142 |
| 3 | 15.950050 | 15 | 15.970841 | 27 | 16.132560 |
| 4 | 15.964984 | 16 | 15.984601 | 28 | 16.157781 |
| 5 | 15.955090 | 17 | 16.000762 | 29 | 16.169298 |
| 6 | 15.975818 | 18 | 16.023135 | 30 | 16.193631 |
| 7 | 15.985813 | 19 | 16.043425 | 31 | 16.214059 |
| 8 | 16.000328 | 20 | 16.063904 | 32 | 16.206200 |
| 9 | 15.986551 | 21 | 16.082811 | 33 | 16.167337 |
| 10 | 15.968143 | 22 | 16.097283 | 34 | 16.178998 |
| 11 | 15.981789 | 23 | 16.114937 | 35 | 16.197996 |
| 12 | 15.948540 | 24 | 16.120712 | 36 | 16.213181 |

Table 2: Lag selection based on AIC

While AIC is the smallest for two lags, it does not follow the identifying assumption justified by Kilian, that is, oil supply takes a longer time to adjust and quarterly shocks are predetermined. Furthermore, using 24 lags proves to be adequate as it has a comparable AIC to the minimum AIC score, justifying our model selection.

| Series | P value |
|---|---|
| $\Delta prod$ | 0.999981 |
| rea | 0.767465 |
| rpo | 1.000000 |

Table 3: Ljung-Box Test, $H_0$ = model residuals are i.i.d.

Ljung-Box Test fails to reject the null hypothesis, indicating to evidence for lag 24 model is adequate.

## 7.2 MCMC Diagnostics

Trace plots and running posterior means of selected coefficients and innovation variances indicate that the chain stabilized after a burn-in of 2,000 iterations. The chain fluctuates around the posterior mean without visible and systematic trends. The full Gibbs sampler completed in under 10 seconds.

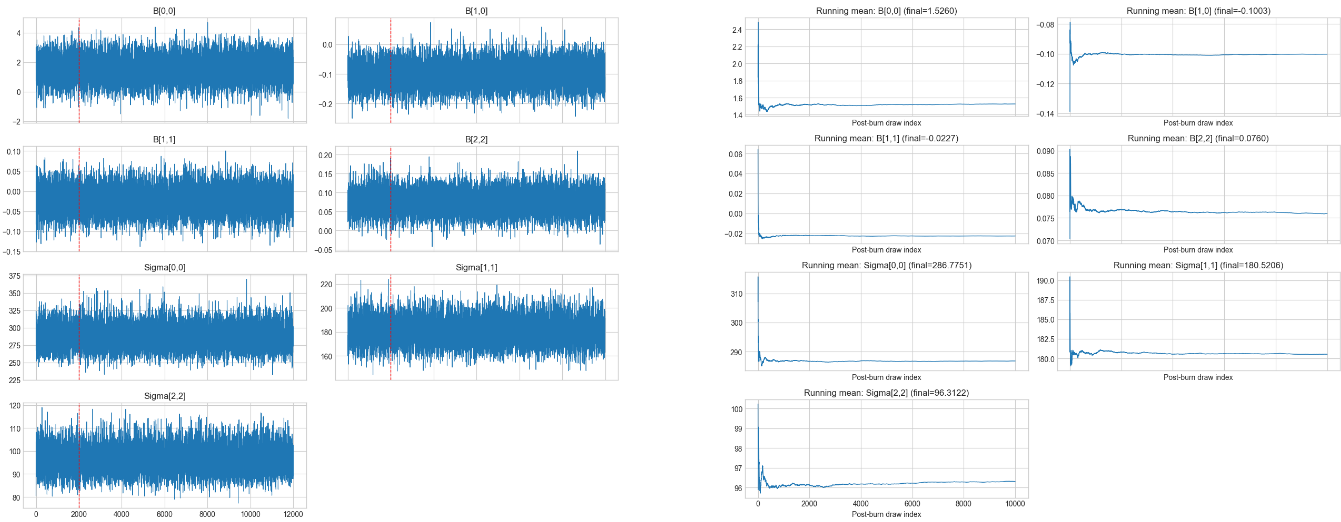


Figure 7: Trace plot and running posterior mean of intercept and lag-1 oil production coefficient, and reduced-form innovation

## 7.3 Posterior Estimates

Table 4 shows summaries of the posterior estimates. The full table is truncated at lag 1 because of length constraints.

Table 4: Posterior summaries of reduced-form BVAR(24) parameters (NIW prior, Gibbs sampler), structural-form residuals

| Equation | Regressor | Mean | 95% credible interval |
|---|---|---|---|
| *Intercept* | | | |
| Oil production | Constant | 1.526 | [-0.107, 3.168] |
| Real activity | Constant | 0.472 | [-0.835, 1.773] |
| Real oil price | Constant | -0.547 | [-1.477, 0.410] |
| *Lag-1 coefficients* | | | |
| Oil production | $\Delta prod_{t-1}$ | -0.100 | [-0.180, -0.021] |
| Oil production | $rea_{t-1}$ | 0.011 | [-0.089, 0.110] |
| Oil production | $rpo_{t-1}$ | 0.081 | [-0.057, 0.218] |
| Real activity | $\Delta prod_{t-1}$ | -0.023 | [-0.085, 0.040] |
| Real activity | $rea_{t-1}$ | 1.076 | [0.998, 1.152] |
| Real activity | $rpo_{t-1}$ | 0.256 | [0.146, 0.369] |
| Real oil price | $\Delta prod_{t-1}$ | -0.036 | [-0.081, 0.010] |
| Real oil price | $rea_{t-1}$ | 0.076 | [0.019, 0.133] |
| Real oil price | $rpo_{t-1}$ | 1.251 | [1.171, 1.331] |
| *Structural innovation covariance* $\mathrm{Cov}(\varepsilon_t)$ | | | |
| oil supply shock | $\Sigma_{11}^{(s)}$ | 1.117 | [0.999, 1.241] |
| commodity agg. demand shock | $\Sigma_{21}^{(s)}$ | -0.004 | [-0.073, 0.063] |
| commodity agg. demand shock | $\Sigma_{22}^{(s)}$ | 1.112 | [0.993, 1.239] |
| oil specific demand shock | $\Sigma_{33}^{(s)}$ | 1.157 | [1.042, 1.276] |
| oil specific demand shock | $\Sigma_{31}^{(s)}$ | -0.049 | [-0.099, 0.004] |
| oil specific demand shock | $\Sigma_{32}^{(s)}$ | -0.039 | [-0.102, 0.026] |

Based on 10,000 post-burn-in draws (burn-in = 2,000, total = 12,000). $\Sigma_{ij}^{(s)}$ denotes the $(i, j)$ element of the sample covariance of $\hat{\varepsilon}_t$ at each Gibbs draw; $\hat{\varepsilon}_t = \hat{B}_0^{-1}\hat{u}_t$ with $\hat{B}_0 = \mathrm{chol}(\hat{\Sigma})^T$, where $\hat{\Sigma}$ denotes the sample covariance of reduced-form innovation.

### 7.4 Robustness: Granger Causality

The Granger causality test shows that the real oil price can be predicted by lagged value of the other variables.

Table 5: Multivariate Granger causality tests at lag $p = 24$ (VAR system). $H_0$: lags of the cause do not help predict all other endogenous variables.

| Cause (other variable) | $F$ | $p$-value |
|---|---|---|
| oil production ($\Delta prod$) | 0.476 | 0.9992 |
| real activity (rea) | 0.875 | 0.7133 |
| real price of oil (rpo) | 1.996 | 0.0001*** |

### 7.5 Summary of Historical Event in Oil Market


Figure 8: Time series plot of the data: 1973-2025

| Year | Event |
|---|---|
| 1973 | October War and Oil Embargo |
| 1980 | Iranian Revolution |
| 1981 | Iran-Iraq War |
| 1990 | Invasion of Kuwait |
| 2001 | OPEC Meeting |
| 2002 | Venezuela civil unrest |
| 2003 | Iraq War |
| 2008–09 | Financial Crisis |
| 2010–12 | Arab Spring |
| 2014–16 | Shale Boom |
| 2015–16 | Global Slowdown |
| 2020 | COVID-19 Pandemic |
| 2022 | Russia–Ukraine War |

Table 6: Major oil market events in the sample period